

Part C – Opinion

Problem: Answers are outdated or irrelevant after 6 months despite correct source documents.

Investigation Strategy

1. Metadata and Sink Validation (Staleness)

Verify whether the Ingestion Pipeline successfully committed the latest updates to Chroma. In on-premises environments, pipelines may finish without error while failing to overwrite existing records.

Method: Conduct a Metadata Audit by querying Chroma for the document_hash or timestamp of recently updated files. If these match previous versions, the Embedding Service or Chroma upsert logic is failing to replace stale vectors.

2. Retrieval Precision and Index Drift (Irrelevance)

If the index is current but answers remain irrelevant, the semantic link between queries and new documentation is likely weak.

Method: Audit Top K Retrieval via the RAG application and log the Cosine Similarity scores. Low scores indicate that the embedding model cannot adequately represent new technical jargon. In this case, investigate implementing Hybrid Search (BM25 + Vector) to ensure specific keywords trigger the correct context.

3. Grounding and Context Adherence (Hallucination)

If retrieval is accurate, but the output is incorrect, the vLLM Inference layer is likely ignoring the provided context.

Method: Perform a Grounding Test by manually injecting correct Chroma chunks into a prompt. If the LLM still provides outdated information, edit the System Prompt with stricter extractive constraints and verify that the vLLM context window is not truncating the retrieved data.